

AMAZON PRODUCT ANALYTICS PROJECT

1. Project Overview

The project is an end-to-end Amazon Product Analytics study. The objective was to convert a raw Amazon product dataset into a structured analytical dataset and then use Python, MySQL, sentiment analysis, and Excel to examine product pricing, discounts, ratings, review volume, product categories, and review sentiment.

The submitted project files show four major analytical stages: data cleaning and feature preparation, SQL-based business analysis, review sentiment analysis, and an Excel-based analytical dashboard.

2. Data Used

Item	Details
Raw dataset	amazon.csv — 1,465 product records, 16 columns
Cleaned dataset	amazon_cleaned.csv — 1,348 product records, 11 columns
Final analytical dataset	amazon_with_sentiment.csv — 1,348 product records, 13 columns
Main categories	9 categories
Primary fields	Product ID, product/category information, actual price, discounted price, discount %, rating, rating count, review text

3. Step 1 — Data Cleaning and Preparation

I used Python with Pandas and NumPy to clean the raw Amazon dataset before performing analysis. The cleaning script converts price, discount, rating, and review-count fields from text formats into numeric values, handles invalid ratings, separates the category hierarchy, removes incomplete analytical records, removes duplicate product IDs, and creates additional analytical variables.

The following transformations were performed:

- Removed the ₹ symbol and comma separators from actual_price and discounted_price and converted them to numeric values.
- Removed the % symbol from discount_percentage and converted it to numeric form.
- Converted rating to numeric form and coerced invalid/non-numeric values to missing values.
- Removed comma separators from rating_count and converted it to numeric form.
- Split the hierarchical category field into main_category and sub_category.
- Removed rows missing discounted price, actual price, rating, or rating count.
- Removed duplicate records using product_id.
- Created savings_amount = actual_price – discounted_price.
- Created price_bucket categories: Under 500, 500–1500, 1500–5000, 5000–20000, and 20000+.
- Created a lean analytical table containing the fields required for SQL and Excel analysis.

The raw dataset contained 1,465 rows and the cleaned dataset contained 1,348 rows. Thus, 117 records were removed during the cleaning/deduplication process.

4. Step 2 — SQL / MySQL Business Analysis

The project includes a MySQL analysis workflow. The cleaned CSV is loaded into a MySQL table named products in the amazon_project database. SQL queries are then used to answer business-oriented questions rather than only performing descriptive statistics.

The SQL analysis covered:

- Top categories using an estimated demand/revenue proxy calculated as $\text{discounted_price} \times \text{rating_count}$.
- Average discount percentage by main category.
- Best-rated categories, with a minimum total review-volume condition to reduce noisy results.
- Products with discounts above 50% but ratings below 3.5 as potential quality red flags.
- Product distribution and average rating across price buckets.
- Highest-rated products with more than 5,000 reviews.

5. Key Analytical Findings

Across the final dataset, the average product rating was 4.09, the average listed discount was 46.7%, and the dataset contained 23,801,431 recorded reviews across the products.

5.1 Category-level demand proxy

The SQL query uses discounted price $\times$ rating count as an estimated demand/revenue proxy. This is not actual Amazon sales revenue; it is an analytical proxy based on the available price and review-volume fields.

Main Category	Products	Estimated Demand/Revenue Proxy
Electronics	490.0	₹56,620,575,045
Home&Kitchen	447.0	₹6,262,253,968
Computers&Accessories	373.0	₹5,955,036,900
MusicalInstruments	2.0	₹64,458,076
OfficeProducts	31.0	₹45,786,730

5.2 Average discount by category

Main Category	Average Discount	Products
HomeImprovement	57.5%	2
Computers&Accessories	53.1%	373
Health&PersonalCare	53.0%	1
Electronics	49.9%	490
MusicalInstruments	46.0%	2

5.3 Best-rated categories

Main Category	Average Rating	Total Reviews
OfficeProducts	4.31	149,675
Toys&Games	4.30	15,867
HomeImprovement	4.25	8,566
Computers&Accessories	4.15	6,335,177
Electronics	4.08	14,208,406

5.4 Price bucket analysis

Price Bucket	Products	Average Rating
Under 500	499	4.07
500-1500	400	4.08

1500-5000	246	4.09
5000-20000	152	4.15
20000+	51	4.24

6. Step 3 — Review Sentiment Analysis

I performed text-based sentiment analysis on the review_content field using TextBlob. For each product, the review text was converted into a sentiment polarity score ranging from -1 (negative) to +1 (positive). The score was then converted into Positive, Neutral, or Negative labels using the thresholds defined in the submitted sentiment.py script.

The sentiment results were merged back with the cleaned product table using product_id, creating the final amazon_with_sentiment.csv dataset.

Sentiment Label	Products	Average Star Rating
Positive	1,239	4.11
Neutral	96	3.89
Negative	13	3.95

The results show that most products in the final dataset were classified as Positive. The project also compares average star ratings across sentiment groups and identifies categories containing the highest number of negative-sentiment products.

Category	Negative-Sentiment Products
Electronics	5
Home&Kitchen	4
Computers&Accessories	3
OfficeProducts	1

7. Step 4 — Excel Dashboard

I created Amazon_Product_Analytics_Dashboard.xlsx using Python and openpyxl. The workbook contains separate Data, Summary, and Sentiment sheets.

- Data sheet containing the final analytical dataset in an Excel Table with filtering and sorting.
- Summary sheet with category-level KPIs including product count, average discounted price, average discount percentage, average rating, and total reviews.
- Overall KPI section showing total products, average rating, average discount, and total reviews.
- Formula-driven Excel analysis using COUNTIF, AVERAGEIF/AVERAGEIFS-style logic, and SUMIF/SUMIFS-style logic.
- A bar chart showing average rating by category.
- A pie chart showing product count by category.
- Sentiment sheet comparing Positive, Neutral, and Negative product counts and their average star ratings.

8. Business Questions Addressed

- Which product categories have the largest estimated demand/revenue proxy?
- Which categories offer the highest average discounts?
- Which categories have the strongest average ratings while maintaining meaningful review volume?

- Which heavily discounted products may have relatively weak customer ratings?
- How are products distributed across different price ranges?
- Which highly rated products also have strong review volumes?
- Does the sentiment of review text broadly correspond with numerical star ratings?
- Which categories contain more negative-sentiment products?
- How can these findings be presented in a management-friendly Excel dashboard?

9. Tools and Technologies Used

Tool / Technology	Purpose
Python	Overall data-processing and analytics workflow
Pandas	Data loading, cleaning, transformation, grouping and analysis
NumPy	Numerical operations and price-bucket preparation
TextBlob	Review-text sentiment polarity and sentiment labels
MySQL / SQL	Structured storage and business-oriented analytical queries
MySQL connector	Python-to-MySQL data loading and query execution workflow
Excel / openpyxl	Dashboard workbook, formulas, tables and charts

10. Project Workflow

Raw Amazon Dataset → Python Data Cleaning → Cleaned Dataset → MySQL/SQL Business Analysis + TextBlob Sentiment Analysis → Final Analytical Dataset → Excel Dashboard

11. Conclusion

The project demonstrates an end-to-end product analytics workflow rather than only a single analysis. I started with raw product and review data, cleaned and transformed the dataset, engineered analytical variables, performed SQL-based business analysis, added review sentiment analysis, and finally converted the results into an Excel dashboard. The final output provides a structured view of pricing, discounts, ratings, review volume, categories, price ranges, and customer-review sentiment.